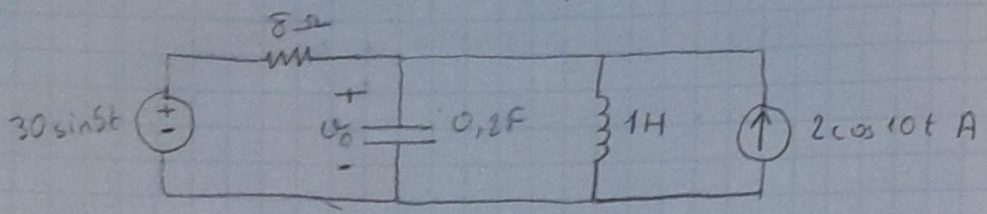
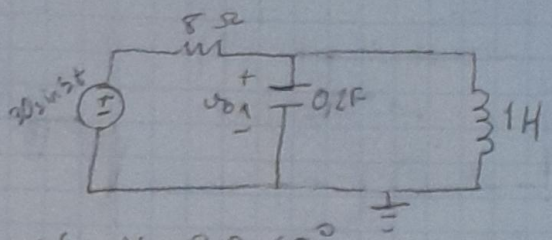
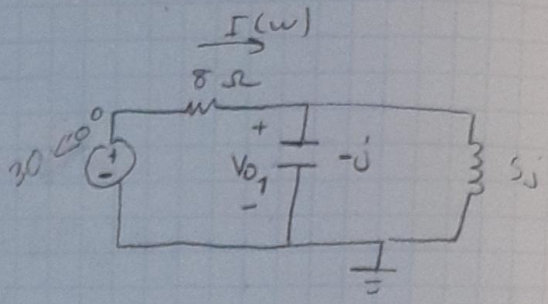
ELEC 226 Homework 1

Q1) Calculate v_o voltage in the circuit by using Superposition (Use the given notation for v_o voltage)



a)





w=5 $V_s = 30\angle 0°$

$Z_C = \frac{1}{jwC} = \frac{1}{j5 \cdot 0,2} = -j$

$Z_L = jwL = j5 \cdot 1 = 5j$

$Z_{eq_1} = 8 + \frac{-j \cdot 5j}{-j+5j} = 8 + \frac{5}{4j}$

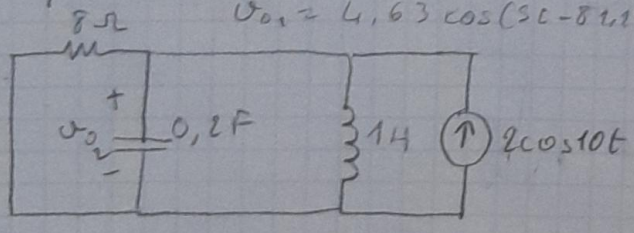
$Z_{eq_1} = 8 - 1,25j$

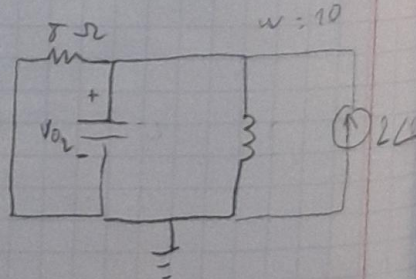
$I(w) = \frac{30\angle 0}{8 - 1,25j} = 3,70 \angle 8,88°$

$V_{o_1} = 30\angle 0° - 8\, I(w) = 4,63 \angle -81,1°$

$v_{o_1} = 4,63 \cos(5t - 81,1°)$

b)





$Z_C = \frac{1}{j10 \cdot 0,2} = \frac{1}{2j} = -0,5j$

$Z_L = jwL = 10j$

$$Z_{eq2} = Z_R // Z_C // Z_L$$

$$= Z_R // \frac{-0{,}5j \cdot 10j}{10j - 0{,}5j} = Z_R // \frac{+5}{9{,}5j}$$

$$Z_R // -0{,}53j$$

$$= \frac{8(-0{,}53j)}{8 - 0{,}53j} = 0{,}529 \angle -76{,}2°$$

$$V_{o2} = I(w)\, Z_{eq2}(w) = (2 \angle 0°)(0{,}529 \angle -76{,}2°)$$

$$= 1{,}058 \angle -76{,}2°$$

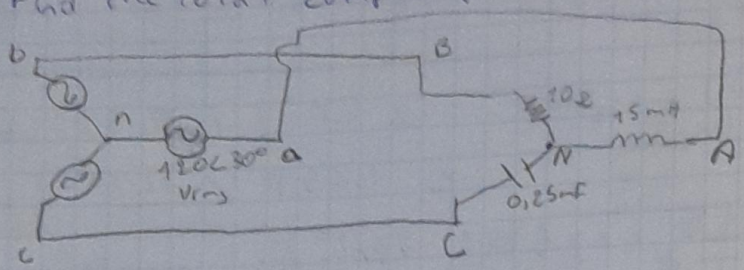
$$v_{o2} = 1{,}058 \cos(10t - 86{,}2°)$$

$$v_o = v_{o1} + v_{o2} = 4{,}63 \cos(5t - 81{,}1°) + 1{,}058 \cos(10t - 86{,}2°)$$

Q-2) For the 3 ph system given below, assume a balanced source with a positive phase sequence, if the operating frequency is 50 Hz

a) Find the magnitude of V_{AN}, V_{BN}, V_{CN} voltages.

b) Find the total complex power supplied by the sources



$V_{AN} = 120\angle 30^\circ$

$V_{BN} = 120\angle 30 \; \angle -120^\circ = 120\angle -90^\circ$

$V_{CN} = 120\angle 30^\circ \angle 120^\circ = 120\angle 150^\circ$

$|V_{AN}| = |V_{BN}| = |V_{CN}| = \boxed{120 \text{ V rms}} = 120\sqrt{2} = 169,7$ formun

$w = 2\pi f = 314,16$

b) $I_a = \frac{V_{an}}{Z_L} = \frac{V_{an}}{j(314,16)10^{-3}} = \frac{120\angle 30^\circ}{j(314,16)10^{-3}} = 381,97\angle -60^\circ$

$S_a = V_{AN} I_A^* = (120\angle 30^\circ)(381,97\angle 60^\circ)$

$= 45836,4\angle 90^\circ = 45836,4j$ VA

$I_b = \frac{V_{BN}}{Z_R} = \frac{120\angle -90^\circ}{10} = 12\angle -90^\circ$

$S_b = V_{BN} I_B^* = 120\angle -90^\circ \; 12\angle +90^\circ = 1440$ VA

$I_c = \frac{V_{CN}}{Z_C} = \frac{120\angle 150^\circ}{\frac{1}{j(314,16).0,25.10^{-3}}} = \frac{120\angle 150^\circ}{12,73\angle -90^\circ}$

$= 9,42\angle -120^\circ$

$S_C = V_{CN} I_C^* = (120\angle 150^\circ)(9,42\angle +120^\circ) = 1130,4\angle -90$

$= -1130,4j$

$S_{delivered} = S_A + S_B + S_C = 45836,4j + 1440 - 1130,4j$

$= 1440 + 44706j$

Q3) For the circuit, unknown values are:

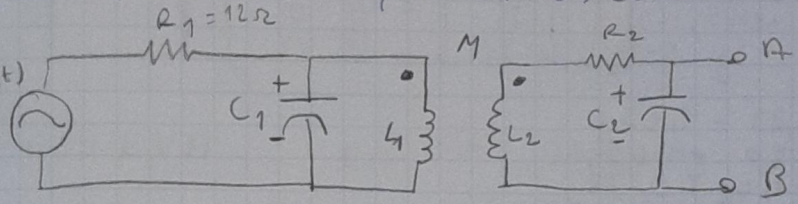
$V_s(t) = 100\cos(5t)$, $R_1 = 12\,\Omega$, $R_2 = 5\,\Omega$, $L_1 = 0{,}3\,H$,

$L_2 = 1{,}2\,H$, $C_1 = C_2 = \frac{1}{100}\,F$ and $M = 0{,}6\,H$

a) Calculate the Thevenin Equivalent circuit between A-B port.

b) What is the impedance value which satisfy the max power between A-B.

c) For the impedance value found in b, calculate the apparent power.

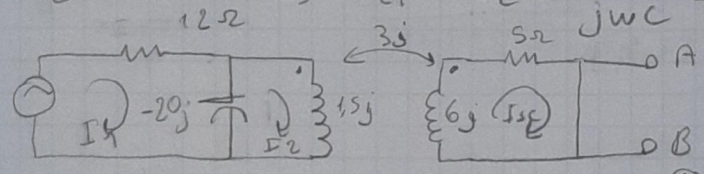


$Z_{L_1} = j\omega L_1 = j \cdot 5 \cdot 0{,}3 = 1{,}5j\,\Omega$

$Z_{L_2} = j\omega L_2 = j \cdot 5 \cdot 1{,}2 = 6j\,\Omega$

$C_1 = C_2 = 10^{-2}\,F \Rightarrow Z_{C_1} = Z_{C_2} = \frac{1}{j\omega C} = \frac{1}{j\,5 \cdot 10^{-2}} = -20j\,\Omega$

1. Isc'yi bulalım



$-100 + 12I_1 - 20jI_1 + 20jI_2 = 0$ ① $\quad -100 + (12 - 20j)I_1 + 20jI_2 = 0$

$I_2(-20j + 1{,}5j) + 20jI_1 - 3jI_{sc} = 0$

② $\quad -18{,}5jI_2 + 20jI_1 - 3jI_{sc} = 0$

③ $\quad (6j + 5)I_{sc} - 3jI_2 = 0$

$$\begin{matrix} & I_1 & I_2 & I_{sc} \end{matrix}$$

$$\begin{bmatrix} (12-20j) & 20j & 0 \\ 20j & -18{,}5j & -3j \\ 0 & -3j & 6j+5 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_{sc} \end{bmatrix} = \begin{bmatrix} 100 \\ 0 \\ 0 \end{bmatrix}$$

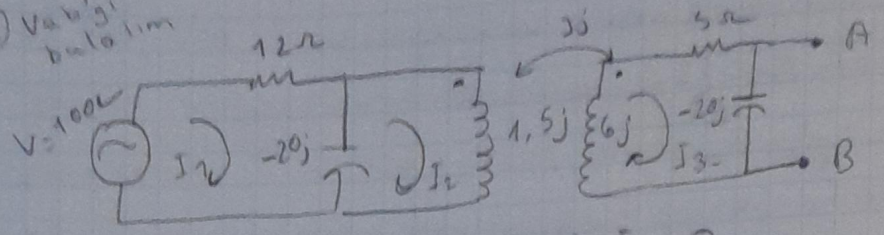
Bunu çözerek bunları elde edeceiz

$I_1 = 7{,}6 - 0{,}36j$

$I_2 = 8{,}02 - 0{,}67j$

$I_{sc} = 2{,}53 + 1{,}77j$

(2) Vaibj bulalim



1) $-100 + (12 - 20j) I_1 + 20j I_2 = 0$

2) $I_2 (-18,5) + 20j I_1 - 3j I_3 = 0$

3) $\left(\frac{6j + 5 \cdot (-20j)}{5 - 10j}\right) I_3 - 3j I_2 = 0$

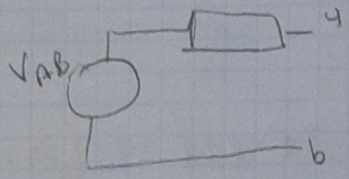
$$\begin{bmatrix} 12-20j & 20j & 0 \\ 20j & -18,5j & -3j \\ 0 & -3j & 5-16j \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 100 \\ 0 \\ 0 \end{bmatrix}$$

$I_1 = 7,93 - 1,45j$
$I_2 = 8,80 - 1,69j$
$I_3 = -1,41 + 0,76j$

$V_{AB} = I_3 \cdot (-20j) = (-1,41 + 0,76j)$

$= 15,2 + 28,2j$

$Z_{Th} = \frac{V_{AB}}{I_{sc}} = \frac{15,2 + 28,2j}{2,53 + 1,77j} = 9,27 + 4,66j$



b) For max power transfer Z_L may be equal to

$Z_{Th}^* = 9,27 - 4,66j$ $\quad V = I \cdot R \quad I = \frac{V}{R}$

c) $S = \frac{V I_2^*}{2} = \frac{V_{AB}}{2} \cdot \left(\frac{V_{AB}}{Z_{Th} + Z_{Th}^*}\right)^*$

$= 27,68 - 0j$

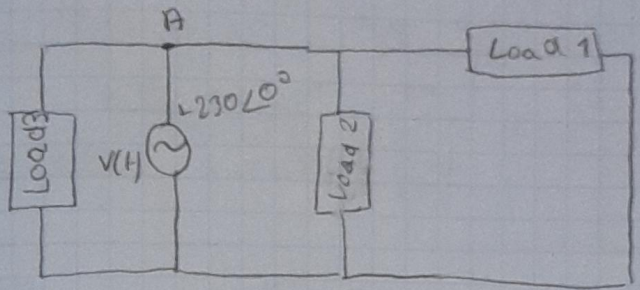
$|S| = \sqrt{(27,68)^2} = 27,68$ VA

Q-4) The source $V(t) = 230\angle 0°$ feeds 3 parallel Loads. For the loads below

Load 1 : $S = 25\ kVA$, $PF = 0,6$ inductive,
Load 2 : $P = 5kW$, $PF = 0,5$ capacitive
Load 3 : $P = 10kW$, $PF = 1$

a) Calculate the condansator value to set the power factor at 0,95 (Freq = 50 Hz)

b) Find the current value before and after the compansation. Explain the difference, if it exists



$|S_1| = 25\ kVA$ $\quad PF_1 = 0,6$ lagging

$\theta_1 = \arccos(0,6) = 53,13°$ $\quad = \theta_V - \theta_I$

$S_1 = 25\ \angle 53,13°\ kVA$

$P_2 = 5kW$ $\quad P_{f2} = 0,5$ leading

$\theta_2 = -\arccos(0,5) = -36,87°$

$S_2 = P_2 + j\,Q_2 = \frac{V_m I_m}{2}\cos(\theta_2) + j\,\frac{V_m I_m}{2}\sin(\theta_2)$

$\frac{V_m I_m}{2} = \frac{P_2}{0,8} = \frac{5kW}{0,8} = 6,25\ kW$

$\Rightarrow Q_2 = 6,25k \cdot \sin(-36,87°) = -3,75\ kVAR$

$S_2 = (5 - j3,75)\ kVA = 6,25\ \angle -36,86°\ kVA$

$P_3 = 10kW$ $\quad PF = 1$ so resistive and $S_3 = P_3 = 10kVA$

$S_a = S_0 = S_1 + S_2 + S_3 = 25\ \angle 53,13° + 6,25\ \angle -36,86° + 10$
$= 34,1\ \angle 28,4°\ kVA = 30 + 16,21j\ kVA$

$PF = \cos(\theta_V - \theta_I)$
$= \cos(28,4) = 0,88$

$Q_0 = 16,21\ kVAR$ $\quad P_0 = 30\ kW$ $\quad \theta_0 = 28,4°$

$S_0 = \frac{P_0}{\cos\theta_0} = 34kVA$ $\quad Q_0 = |S_0| \sin\theta_0$

$PF = 0,95 \Rightarrow$ P not changed but the apparent power has changed
$\theta_n = \cos^{-1}(0,95) = 18,19°$

$|S_n| = \frac{P_0}{\cos\theta_n} = \frac{30kW}{0,95} = 31,58\ kVA$ and $Q_n = |S_n| \sin\theta_n = 9,86\ kVAR$

$Q_c = Q_0 - Q_n = 16,21 - 9,86 = 6,35\ kVAR$

$Q_c = 6,35 \text{ kVA}_r$

$$C = \frac{Q_c}{\omega V_{rms}^2} = \frac{6,35k}{2\pi \cdot 50 \cdot 230^2} = 38,31 \mu F$$

b) $S_0 = (30 + j16,21) \text{ kVA} = 34,1 \angle 28,4° \text{ kVA}$

$S_0 = V_0 I_0^*$ $= 230 \angle 0° \quad I_0^* = 34,1k \angle 28,4°$

V_0 and I_0 are rms values

$$I_0^* = \frac{34,1k \angle 28,4°}{230}$$

$$I_0 = 148,2 \angle -28,4° \text{ A}$$

$S_n = (30 + j\,9,86) \text{ kVA} = 31,58 \angle 18,19° \text{ kVA}$

$S_n = V_0 I_n^* = 230 \angle 0° \quad I_n^* = 31,58 \angle 18,19° k$

$$I_n^* = \frac{31,58}{230} \angle 18,19°$$

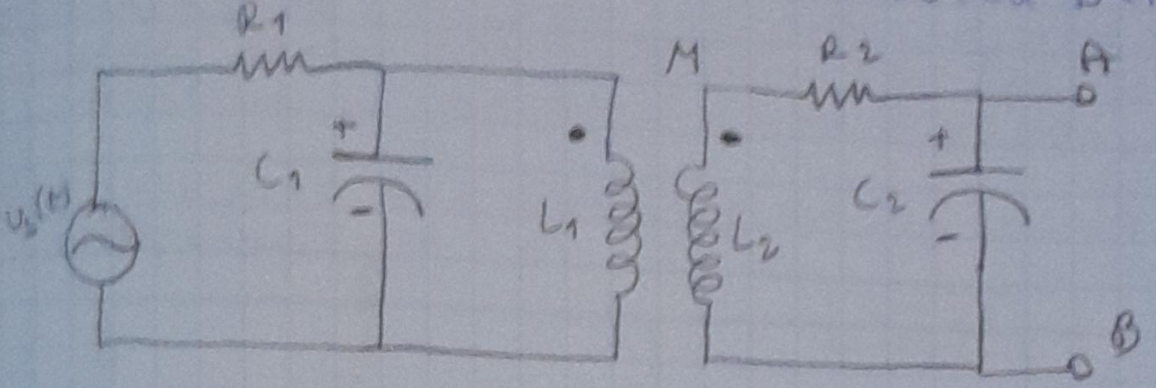
$$I_n = 137,3 \angle -18,19°$$

Adding a capacitor make the current more lagging. Because of increse the capacitance. And the reduction in $|I_n|$ indicative of a decrease in power.

Q2) Voltage source $V_s(t) = 100\cos(5t)$ and $R_1 = 12\Omega$, $R_2 = 3\Omega$, $L_1 = 0,3H$, $L_2 = 1,2H$, $C_1 = C_2 = 1/100F$ and mutual inductance M is $0,6H$.

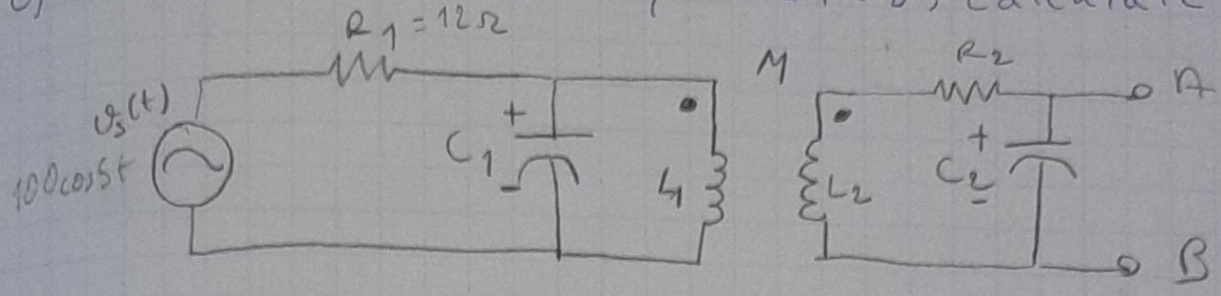
A) Please find thevenin equivalent between A and B

B) To draw maximum power from voltage source, please find the impedance that must be connected between A and B



Bu sorunun cevabı soru B ile aynıdır ve aşağıdaki gibi gösterilmiştir.

c) ... found in b, calculate the apparent power.

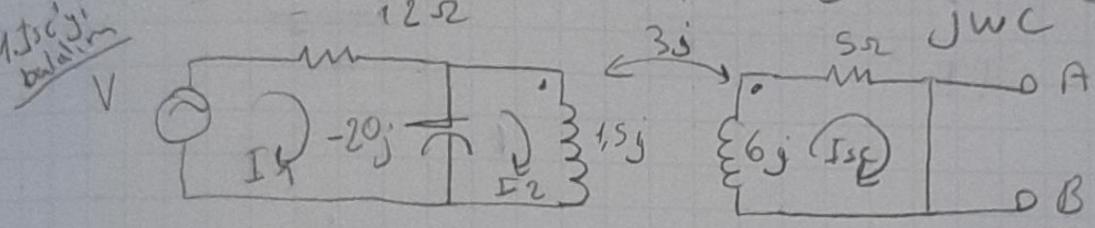


$Z_{L_1} = j\omega L_1 = j \cdot 5 \cdot 0,3 = 1,5j\ \Omega$

$Z_{L_2} = j\omega L_2 = j \cdot 5 \cdot 1,2 = 6j\ \Omega$

$C_1 = C_2 = 10^{-2}\ F \Rightarrow Z_{C_1} = Z_{C_2} = \frac{1}{j\omega C} = \frac{1}{j\,5 \cdot 10^{-2}} = -20j\ \Omega$

1.Isc'yi bulalım



$-100 + 12I_1 - 20jI_1 + 20jI_2 = 0$ (N) $\quad -100 + (12-20j)I_1 + 20jI_2 = 0$

$I_2(-20j + 1,5j) + 20jI_1 - 3jI_{sc} = 0$

(2) $-18,5jI_2 + 20jI_1 - 3jI_{sc} = 0$

(3) $(6j+5)I_{sc} - 3jI_2 = 0$

$$\begin{matrix} & I_1 & I_2 & I_{sc} \end{matrix}$$

$$\begin{bmatrix} (12-20j) & 20j & 0 \\ 20j & -18,5j & -3j \\ 0 & -3j & 6j+5 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_2 \end{bmatrix} = \begin{bmatrix} 100 \\ 0 \\ 0 \end{bmatrix}$$

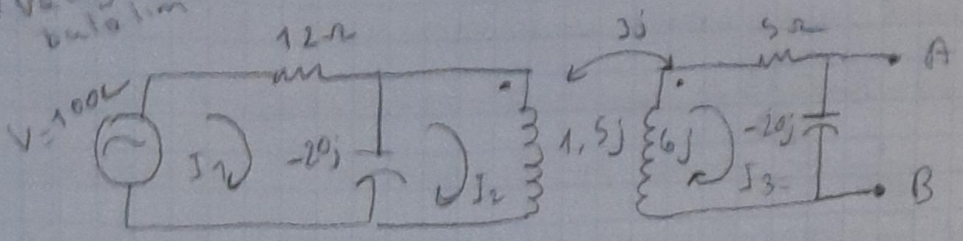
Bunu çözersek bunları elde ederiz

$I_1 = 7,8 - 0,36j$

$I_2 = 8,02 - 0,68j$

$I_{sc} = 2,53 + 1,77j$

② Vedi'ji bulalim



1) $-100 + (12 - 20j) I_1 + 20j I_2 = 0$

2) $I_2 (-17,5) + 20j I_1 - 3j I_3 = 0$

3) $(6j + 5 - 20j) I_3 - 3j I_2 = 0$

$5 - 16j$

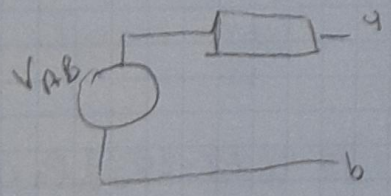
$$\begin{bmatrix} 12-20j & 20j & 0 \\ 20j & -17,5j & -3j \\ 0 & -3j & 5-16j \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 100 \\ 0 \\ 0 \end{bmatrix}$$

$I_1 = 7,93 - 1,45j$
$I_2 = 8,80 - 1,69j$
$I_3 = -1,41 + 0,76j$

$V_{AB} = I_3 \cdot (-20j) = (-1,41 + 0,76j)$

$= 15,2 + 28,2j$

$Z_{Th} = \frac{V_{AB}}{I_{sc}} = \frac{15,2 + 28,2j}{2,53 + 1,77j} = 9,27 + 4,66j$



b) For max power transfer Z_L may be equal to

$Z_L^* = 9,27 - 4,66j$ $\quad V = IR \quad I = \frac{V}{R}$

c) $S = \frac{V I^*}{2} = \frac{V_{AB}}{2} \cdot \left(\frac{V_{AB}}{Z_{Th} + Z_{Th}^*} \right)^*$

$= 27,68 - 0j$

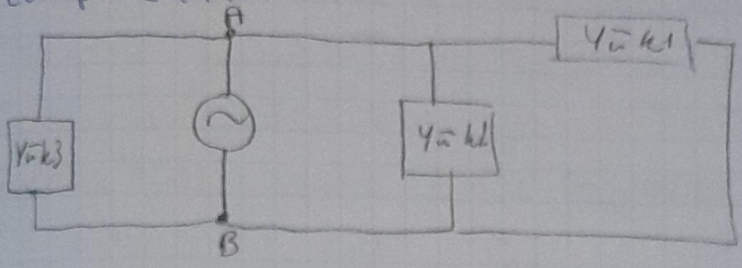
$|S| = \sqrt{(27,68)^2} = 27,68 \text{ VA}$

Q-6) The source whose voltage is 230 V is providing power to 3 loads which are Yük-1, Yük-2 and Yük-3

Yük 1: S = 20 kVA, pf = 0,6 (inductive)
Yük 2: P = 10 kW, pf = 0,8 (capacitive)
Yük 3: P = 10 kW, pf = 0,6 (inductive)

A) Please find the capacity value of the capacitor to make power factor 0,95 (f=50Hz). The location of the capacity must be in between A and B points

B) Please find the current given by source before and after compensation.



$|S_1| = 20\,kVA$ $\quad pf_1 = 0,6$ lagging $\quad \theta_1 = \arccos(0,6) = 53,13° = \theta_y - \theta_{i1}$

$S_1 = 20 \angle 53,13°\ kVA$

$P_2 = 10\,kW$ $\quad pf = 0,8$ leading $\quad \theta_2 = -\arccos(0,8) = -36,87°$

$S_2 = P_2 + jQ_2 = \frac{V_m I_m}{2}\cos(\theta_2) + j\frac{V_m I_m}{2}\sin\theta_2$

$\frac{V_m I_m}{2} = \frac{P_2}{0,8} = \frac{10\,kW}{0,8} = 12,5\,kW$

$\Rightarrow Q_2 = 12,5\,kW \cdot \sin(-36,87°) = -7,50\,kVAR$

$S_2 = (10 - j7,50)\,kVA = 12,5 \angle -36,87°\ kVA$

$P_3 = 10\,kW$ $\quad pf = 0,6$ lagging $\quad \theta_3 = \arccos(0,6) = 53,13°$

$\frac{V_m I_m}{2} = \frac{P_3}{0,6} = \frac{10\,kW}{0,6} = 16,67\,kW$

$Q_3 = 16,67\,kW \cdot \sin(53,13°) = 13,34\,kVAR$

$S_3 = (10 + 16,67j)\,kVA = 19,44 \angle 59,04°$

$S_{ol} = S_o = S_1 + S_2 + S_3 = 20\angle 53,13° + 12,5\angle -36,87 + 19,44\angle 59,04°$
$= 40,71 \angle 38,19° = (\underbrace{32}_{P_o} + j\underbrace{25,17}_{Q_o})\,kVA$

$S_o = \frac{P_o}{\cos\theta_o} = 40,71\,k$ $\quad Q_o = |S_{ol}|\sin\theta_o$

$pf_n = 0,95 \Rightarrow P$ not changed but the apparent power has changed
$\theta_n = \cos^{-1}(0,95) = 18,19°$

$|S_n| = \frac{P_o}{\cos\theta_n} = \frac{32kw}{0,95} = 33,68\ kVA$ and

$Q_n = |S_n| \sin\theta_n = 33,68k \cdot \sin(17,19°) = 10,51\ kVAR$

$S_n = (32 + j\,10,51)kVA = 33,68 \angle 17,19° \ kVA$

$Q_c = Q_o - Q_1$

$= 25,17 - 10,51 = 14,66\ kVAR$

$C = \frac{Q_c}{\omega V^2_{rms}} = \frac{14,66k}{2\pi 50 \cdot 230^2} = 882,21\mu F$

b) $S_o = 40,71 \angle 38,19°\ kVA$

$S_o = V_o I_o^* = 230 \angle 0° \quad I_o^* = 40,71 \angle 38,19°\ kVA$

V_o and I_o are rms values

$I_o^* = 177 \angle 38,19°\ A$

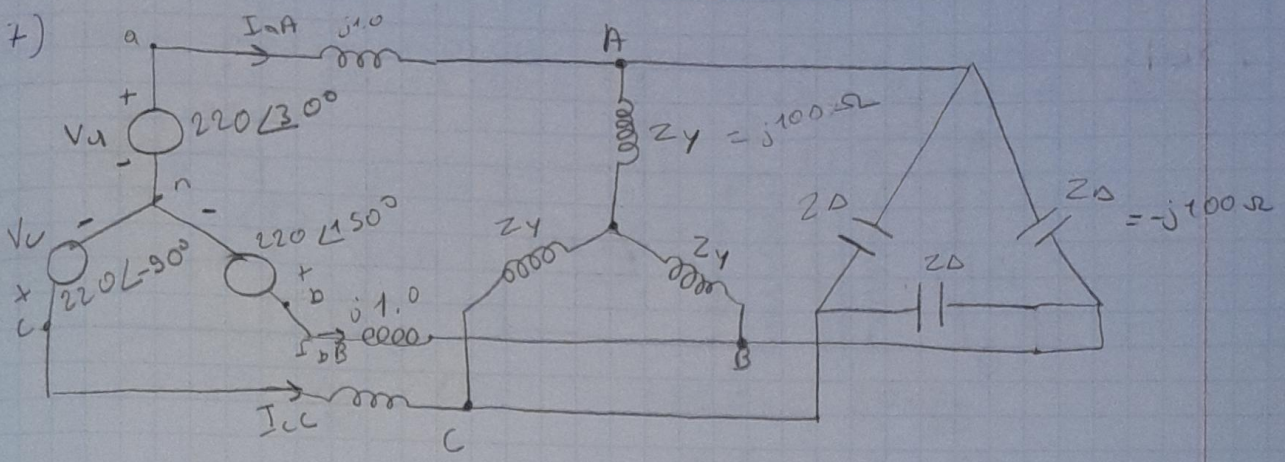
$I_o = 177 \angle -38,19°\ A$

$S_1 = V_o I_n^* = 230 \angle 0° \quad I_n^* = 33,68 \angle 18,18°\ kVA$

$I_n^* = 146,43 \angle 18,18°$

$I_n = 146,43 \angle -18,18°$

Q 7)



Phase voltages are $V_a = 220\angle 30°$, $V_b = 220\angle 150°$ and $V_c = 220\angle -90°$.

The loads information: $Z_\Delta = -j100$ ohm and $Z_Y = j100$ ohm.

A) Please Find I_{aA}, I_{bB} and I_{cC}

B) Please find total active, reactive and apperant power consumed by loads.

Z_Δ'yı 3'e bölersek Y bağlantısındaki Empedans cevirmiş oluruz.

$$Z_{Y_1} = \frac{Z_\Delta}{3} = \frac{-100j}{3}$$

$$Z_{eq} = Z_L + Z_Y \,//\, Z_{Y_1} = Z_L + \frac{Z_Y \cdot Z_{Y_1}}{Z_Y + Z_{Y_1}} = \frac{j100 \cdot \left(\frac{-100j}{3}\right)}{j100 + \left(\frac{-100j}{3}\right)} + Z_L$$

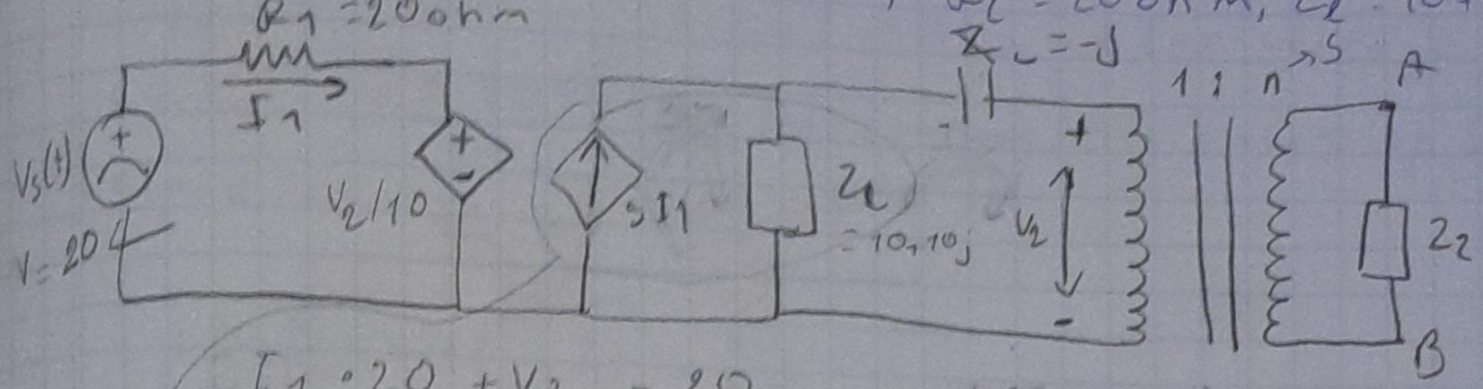
$$= -50j + j = -49j$$

$$I_a = \frac{V_a}{Z_{eq}} = \frac{220\angle 30°}{-49j} = 4{,}44 + 0{,}69j = 4{,}49\ \angle 0{,}15°$$

$$I_b = \frac{V_b}{Z_{eq}} = \frac{220\angle 150°}{-49j} = 4{,}49\ \angle 150{,}15°$$

$$I_c = \frac{V_c}{Z_{eq}} = \frac{220\angle -90}{-49j} = 4{,}49\ \angle -8285°$$

Q 8) Todcow maximumpower from the circuit, please find Z_2 value. $V_s(t) = 20\cos 100t$, $R_1 = 20\,ohm$, $Z_L = 10+10j$, $Z_c = -j$, $n=5$

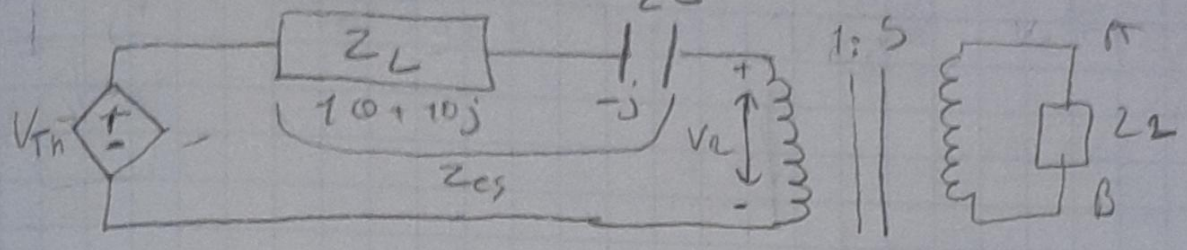


$$I_1 \cdot 20 + \frac{V_2}{10} = 20$$

$$20 I_L = 20 - \frac{V_2}{10}$$

$$5 I_1 = 5 - \frac{V_2}{40}$$

⇒ kaynak dönüşümü yapalım



$$V_{Th} = 5 I_1 \cdot (Z_L) = \left(5 - \frac{V_2}{40}\right)(10+10j)$$

$$Z_{eş} = 10+10j - j = 10+9j$$

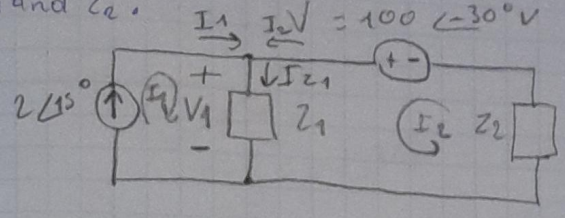
For max power

$$Z_{eş} = \left(\frac{N_1}{N_2}\right)^2 Z_2$$

$$10+9j = \left(\frac{1}{5}\right)^2 = Z_2$$

$$\boxed{250 + 225j = Z_2}$$

Q2) The current source the circuit shown in figure below supplies 131.16 − j36,048 VA and the voltage source supplies 64,2275 − j87,8481 VA. Determine the values of the impedance Z_1 and Z_2.



$$S_1 = \frac{V_1 \cdot I^*}{2} \Rightarrow V_1 = \frac{2S}{I_1^*} = \frac{2(131,16 - j\,36,048)}{(2\angle 15°)^*}$$

$$= \frac{272,05\angle -15,368}{(2\angle -15)} = 136,025\angle -0,368°$$

$$S_2 = \frac{V \cdot I_2^*}{2} \Rightarrow I_2^* = \frac{2S_2}{V} = \frac{2(64,2275 + j\,87,8481)}{100\angle -30}$$

$$= \frac{217,65\angle 53,83°}{100\angle -30} = 2,1765\angle 83,83°$$

$$\Rightarrow I_2 = 2,1765\angle -83,825°$$

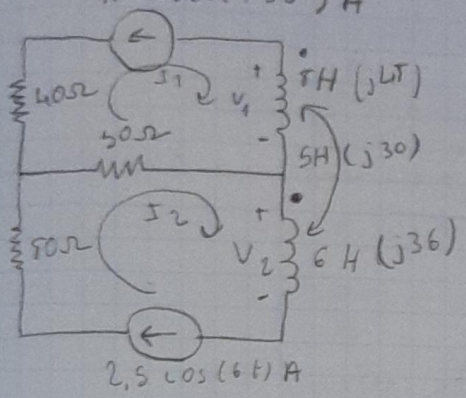
$$Z_1 = \frac{V_1}{\underbrace{I_1 + I_2}_{I_{21}}} = \frac{136,025\angle -0,368°}{2\angle 15° + 2,1765\angle -83,83°} = 50\angle 36,87°$$

$$= 40 + 30j$$

$$Z_2 = \frac{V - V_1}{I_2} = \frac{100\angle -30° - 136,025\angle -0,368°}{2,1765\angle -83,83°}$$

$$= 32,02\angle -51,34° = 20 - 25j$$

Q10) The circuit shown in Figure below is represented in the time domain. Determine coil voltages v1 and v2.



$I_1 = 1,5 \angle -90°$ A

$I_2 = 2,5 \angle 0°$ A olur

$Z_L = jwL$

$V_1 = j48 I_1 + j30 I_2$

$= j48 (1,5 \angle -90°) + j30 (2,5 \angle 0°)$

$= 103,966 \angle 46,17°$ V

$V_2 = j36 I_2 + j30 I_1$

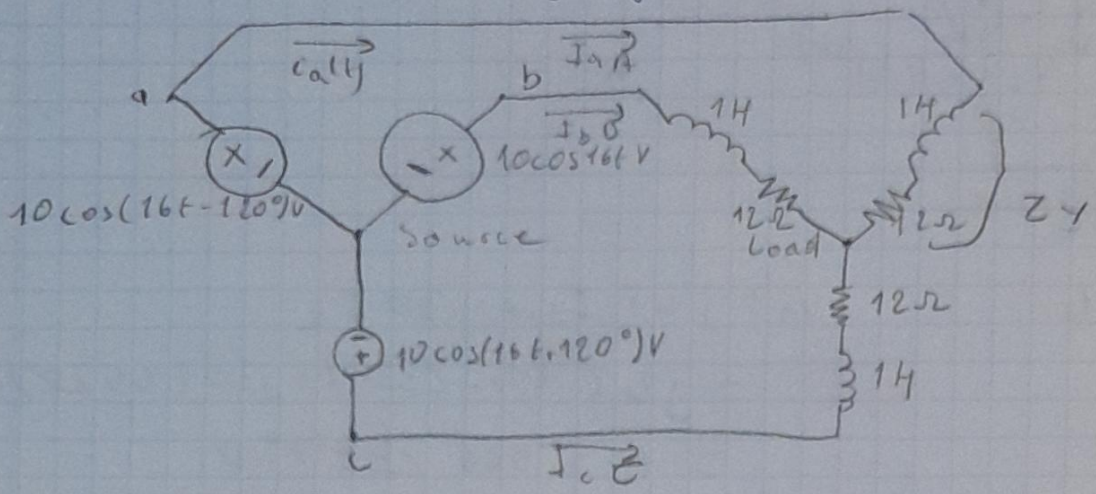
$= j36 (2,5 \angle 0°) + j30 (1,5 \angle -90°)$

$= 100,623 \angle 63,43°$ V

Q-11) Wye connected source and load are shown in Figure below.

a) Determine the rms value of the current $i_a(t)$

b) Determine the average power delivered to the load.



$V_a = 10\angle -120° = 7,07\angle -120° V_{rms}$

$V_b = 7,07\angle 0$ rms

$V_c = 7,07\angle 120$ rms

$Z_Y = 12 + j\omega L = 12 + j16\ \Omega = 20\angle 53,13°$

$I_{aA} = \frac{V_a}{Z_A} = \frac{7,07\angle 0°}{20\angle 53,13°} = 0,3535\angle -53,13$

$= 0,212 - 0,283j$

$I_{bB} = 0,3535\angle -53,13° \angle -120° = 0,3535\angle -173,13°$

$I_{cC} = 0,3535\angle -53,13° \angle -240 = 0,3535\angle 66,87°$

$i_a(t) = 0,3535\cos(16t - 53,13)$

b) $P = 3V_p I_p \cos\theta$

$= 3(7,07)(0,3535)\cos(-53,13)°$

$= 4,499\ W$

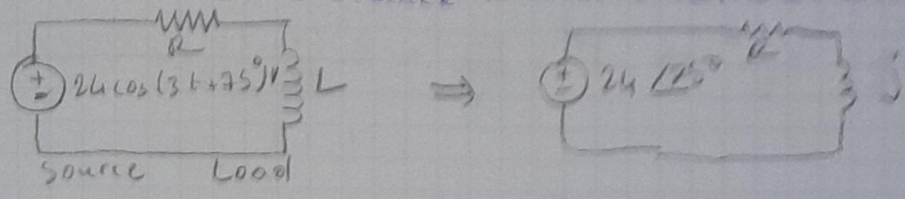
Q12) The circuit in Fig below consists of a source connected to a load.

a) Suppose $R = 9\,\Omega$ and $L = 5$ H. Determine the average, complex, and reactive powers delivered by the source to the load

b) Suppose $R = 15\,\Omega$ and $L = 3$ H. Determine the average, complex and reactive powers delivered by the source to the load.

c) Suppose the source delivers $8,47 + j\,14,1$ VA to the load. Determine the values of the resistance R, and the inductance L.

d) Suppose the source delivers $14,1 + j\,8,47$ VA to the load. Determmine the resistance R and the inductance L.



a) $V = 24\angle 75°$ $\quad Z = R + j\omega L = 9 + j\,15\ \Omega$

$I = \frac{V}{Z} = \frac{24\angle 75°}{9+j15} = 1,37\angle -16°$

Complex power delivered $\rightarrow S = \frac{VI^*}{2} = \frac{24\angle 75° \cdot 1,37\angle -16°}{2} = 16,44\angle 59°$ VA $= 8,47 + 14,1\,j$ VA

Average powerd. $P = 8,47$ W

Reactive power $= Q = 14,1$ VAR

b) $V = 24\angle 75°$ $\quad Z = R + j\omega L = 15 + j\,9\ \Omega$

$I = \frac{V}{Z} = \frac{24\angle 75°}{15 + j\,9} = 1,37\angle 44°$

$S = \frac{VI^*}{2} = \frac{(24\angle 75°)(1,37\angle -44°)}{} = 16,44\angle 31°$ $= 14,05 + 8,46\,j$

Average power $P = 14,05$ W

Reactive power $Q = 8,49$ W

c) $S = 8,47 + j\,14,12$ VA

$S = \frac{VI^*}{2} \Rightarrow I^* = \frac{2S}{V} = \frac{2(8,47 + j\,14,12)}{24 \angle 75°}$

$= 1,37 \angle -15,95$

$\boxed{I = 1,37 \angle 15,95}$

$V = I \cdot Z \qquad Z = \frac{V}{I} = \frac{24 \angle 75}{1,37 \angle 15,95}$

$= 17,52 \angle 59,05$

$Z = 9 + 15j$

$\boxed{R = 9\,\Omega} \qquad \not\Delta\, \omega L = 15\,\not\Omega$

$\downarrow 3$

$\boxed{L = 5H}$

d) $S = 14,12 + j\,8,47$

$S = \frac{VI^*}{2} \Rightarrow I^* = \frac{2S}{V} = \frac{2(14,12 + j\,8,47)}{24 \angle 75°}$

$= 1,37 \angle -44$

$I = 1,37 \angle 44$

$Z = \frac{V}{I} = \frac{24 \angle 75}{1,37 \angle 44} = 17,52 \angle 31°$

$= 15 + 9j$

$R = 15\,\Omega \qquad \not\Delta\, \omega L = 9\,\not\Omega$

$3L = 9$

$L = 3H$